LPG Cylinder Safety & Leakage Analytics

Power BI Dashboard Report

Business Problem

LPG leakage, defective cylinders, delayed complaint resolution, and inspection failures create significant safety, customer service, and cost risks. This Power BI solution provides a centralised view to monitor product quality, identify high-risk zones and plants, track complaints, and improve operational efficiency across three plants — Belagavi, Hubballi, and Mangalore — covering the period January 2024 to May 2026.

Business Value

- Improves safety monitoring across plants and delivery zones
- Identifies high-risk geographic regions for targeted intervention
- Detects problematic cylinder batches and manufacturing runs
- Reduces complaint resolution delays and improves service levels
- Tracks inspection quality and defect rates over time
- Measures financial impact and cost per complaint
- Supports data-driven decision-making at all operational levels

Dashboard Pages

Page	Dashboard	Focus Area
1	Executive Overview	Complaints, costs, inspection & safety metrics
2	Manufacturing Quality Analysis	Inspection outcomes, defects, replacements, plant performance
3	Geographic Risk Analysis	Complaint distribution, high-risk zones, leakage rates
4	Advanced Analytics & Business Impact	Costs, batch analysis, customer segments, risk indicators

Prepared using Microsoft Power BI | Data period: Jan 2024 – May 2026

Page 1 — Executive Overview

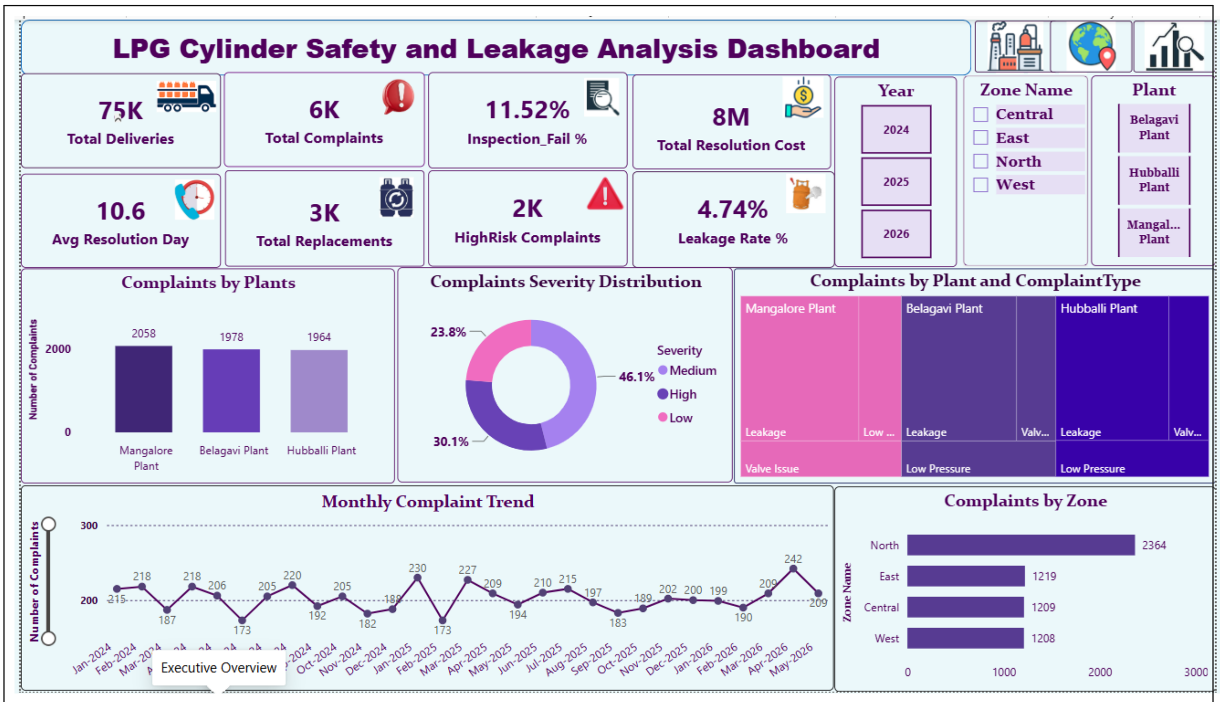

Figure 1 — LPG Cylinder Safety and Leakage Analysis: Executive Overview

Purpose

This is the entry-point dashboard providing stakeholders with a top-level summary of operational health — covering total complaints, leakage rates, resolution costs, and severity distribution across all plants and zones.

Key Metrics

75K	Total Deliveries	**6K**	Total Complaints
4.74%	Leakage Rate %	**2K**	High-Risk Complaints
■8M	Total Resolution Cost	**10.6**	Avg Resolution Days
3K	Total Replacements	**11.52%**	Inspection Fail %

Key Insights

- Mangalore Plant recorded the highest complaints (2,058), with Belagavi (1,978) and Hubballi (1,964) close behind — distribution is broadly even across plants.
- Severity split: 46.1% Medium, 30.1% High, and 23.8% Low — nearly one-third of complaints are high priority requiring urgent action.
- The North zone leads all zones in complaint volume (2,364), making it the primary area for safety inspections and awareness campaigns.
- Monthly complaint volumes average 200+ per month, with peaks in Jun-2025 (230) and Apr-2026 (242), indicating recurring seasonal risk periods.
- Leakage is the dominant complaint type across all three plants — the core safety concern driving costs and replacements.

Page 2 — Manufacturing Quality Analysis

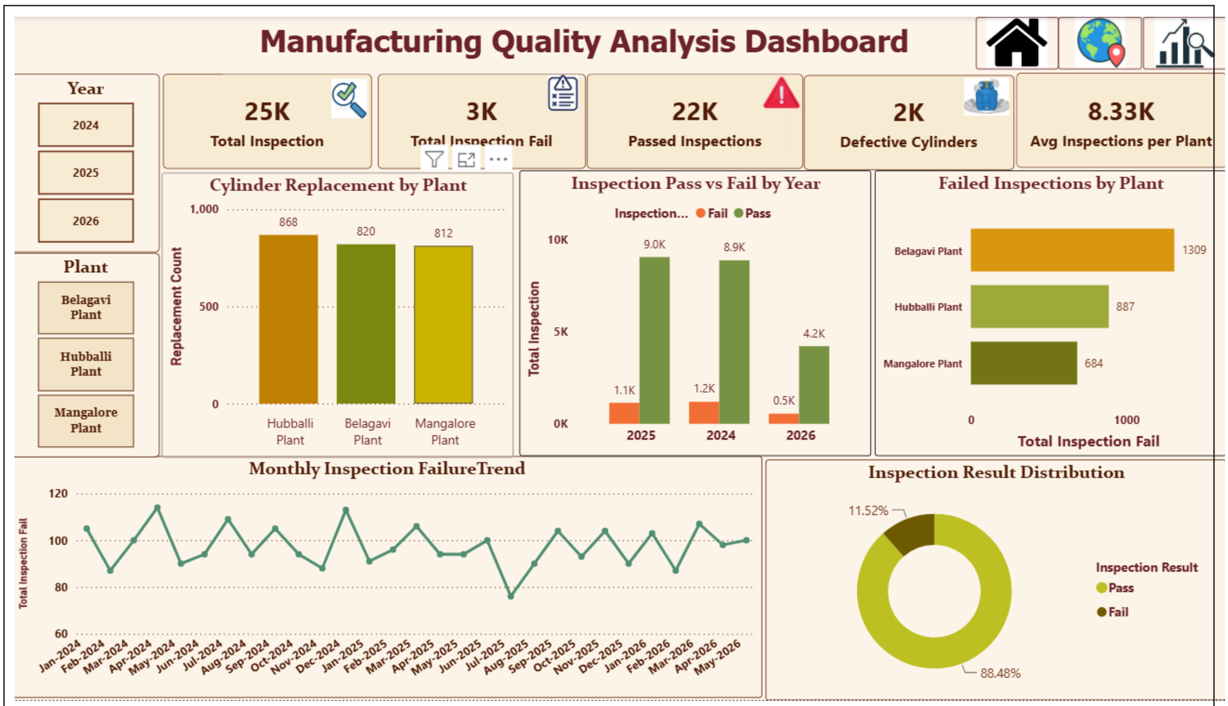

Figure 2 — Manufacturing Quality Analysis Dashboard

Purpose

This dashboard focuses on cylinder inspection outcomes and plant-level quality control, tracking pass/fail rates and replacement volumes to identify manufacturing bottlenecks and defect patterns over time.

Key Metrics

25K	Total Inspections	**3K**	Total Inspection Fails
22K	Passed Inspections	**2K**	Defective Cylinders
88.48%	Inspection Pass Rate	**8.33K**	Avg Inspections / Plant

Key Insights

• An 11.52% inspection failure rate means roughly 1 in 9 cylinders fails quality checks — a systemic issue requiring process-level intervention.

• Belagavi Plant has the highest failed inspections (1,309), followed by Hubballi (887) and Mangalore (684) — Belagavi is the priority plant for quality improvement.

• Cylinder replacement counts are closely matched: Hubballi (868), Belagavi (820), Mangalore (812) — uniform replacement demand across all plants.

• Monthly failure trend oscillates between 80–120 failures per month with no sustained improvement, confirming the need for a structured corrective action plan.

• 2025 recorded the highest inspection volumes (~10K), with 2026 data partial but tracking on a lower trajectory.

Page 3 — Geographic Risk Analysis

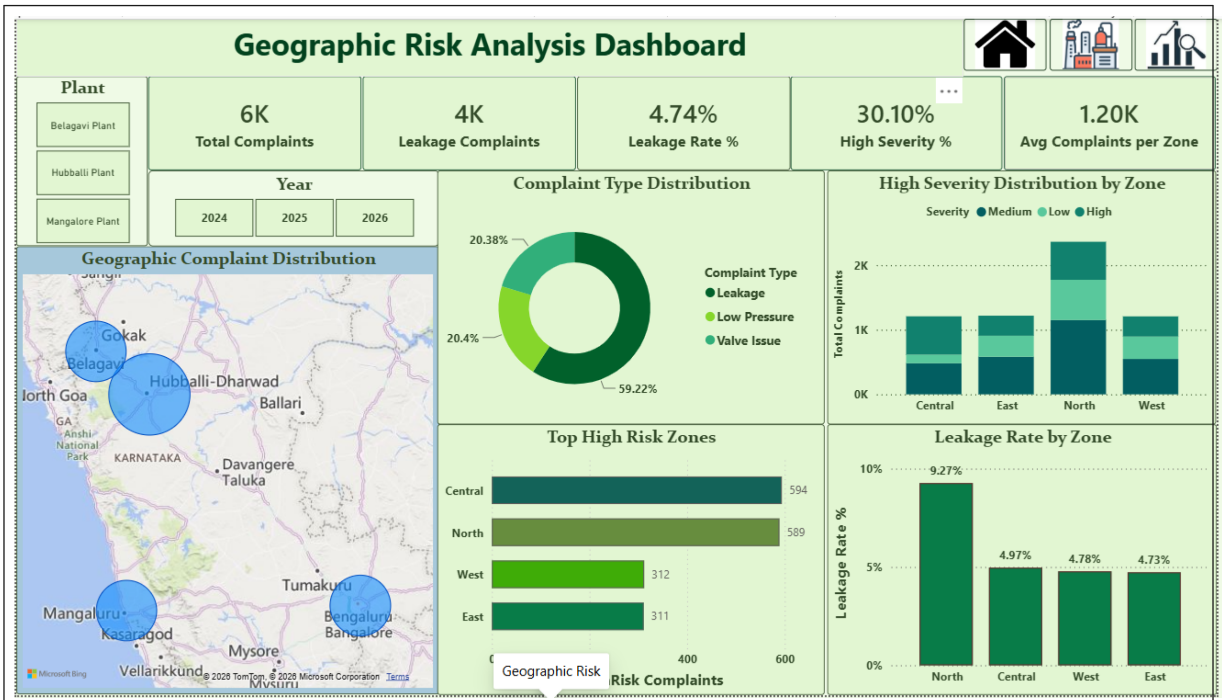

Figure 3 — Geographic Risk Analysis Dashboard with Zone Map

Purpose

This dashboard maps complaint data spatially across Karnataka, enabling zone-level risk identification. It allows field teams and management to pinpoint high-risk delivery areas and prioritise safety inspections geographically.

Key Metrics

6K	Total Complaints	**4K**	Leakage Complaints
4.74%	Leakage Rate %	**30.10%**	High Severity %
1.20K	Avg Complaints / Zone	**9.27%**	North Zone Leakage Rate

Key Insights

- Leakage accounts for 59.22% of all complaint types, confirming it as the primary safety concern — followed by Low Pressure (20.4%) and Valve Issues (20.38%).
- Central (594) and North (589) zones have the highest risk complaint counts and are the top two zones requiring targeted field intervention.
- North zone leakage rate of 9.27% is nearly double all other zones (Central 4.97%, West 4.78%, East 4.73%), making it the single highest-risk area.
- West and East zones show significantly lower complaint volumes (312 and 311 respectively), suggesting better infrastructure or more effective local operations.
- The geographic bubble map confirms Belagavi and Hubballi-Dharwad as the largest complaint-density hubs, enabling visual prioritisation for operations teams.

Page 4 — Advanced Analytics & Business Impact

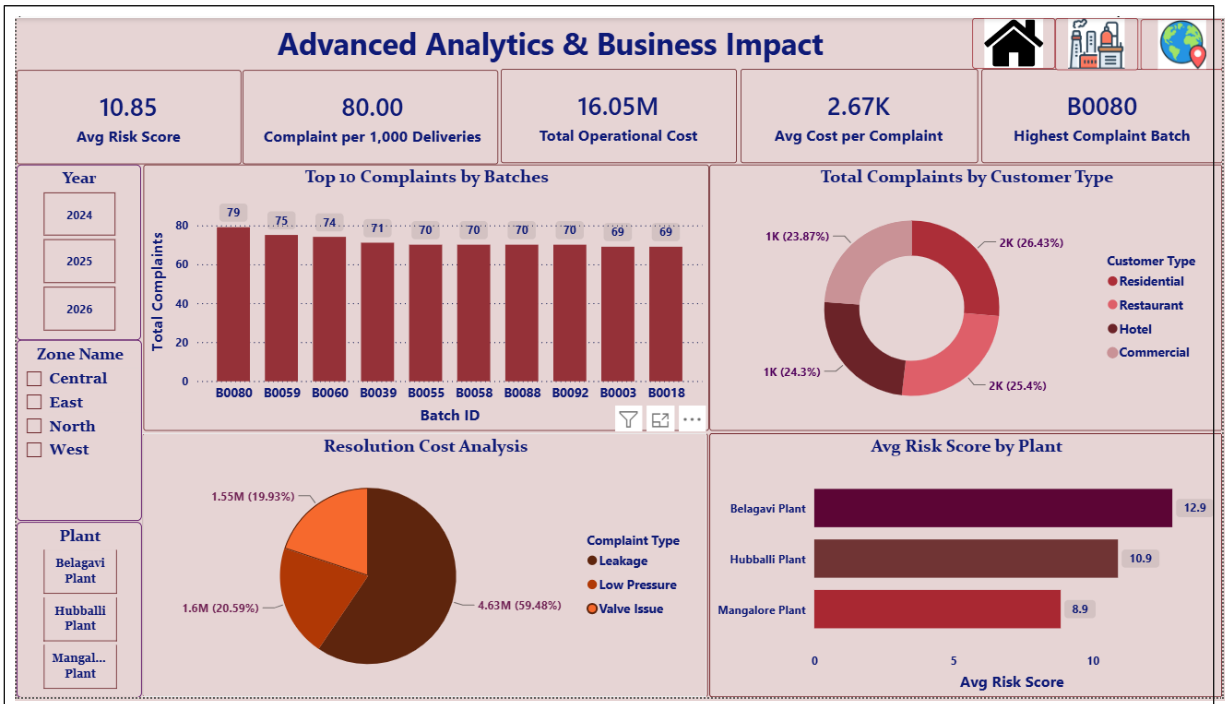

Figure 4 — Advanced Analytics and Business Impact Dashboard

Purpose

This dashboard links operational complaints to financial and strategic metrics — covering resolution costs by complaint type, batch-level quality analysis, customer segment breakdown, and plant-level risk scoring to support data-driven resource allocation.

Key Metrics

■16.05M	Total Operational Cost	**■2.67K**	Avg Cost per Complaint
80	Complaints / 1,000 Deliveries	**10.85**	Avg Risk Score
B0080	Highest Complaint Batch	**12.9**	Belagavi Risk Score

Key Insights

- Leakage complaints generate 59.48% of total resolution costs (■4.63M) — addressing leakage alone could deliver the largest cost savings of any single initiative.
- Low Pressure (■1.6M, 20.59%) and Valve Issues (■1.55M, 19.93%) account for the remaining costs, with both categories also warranting process review.
- Batch B0080 has the highest complaint count (79 complaints), with the top 10 batches ranging from 69–79 — batch-level audits are a key quality intervention point.
- Customer complaints are evenly spread: Residential (26.43%), Commercial (25.4%), Hotel (24.3%), Restaurant (23.87%) — no single segment is disproportionately affected.
- Belagavi Plant carries the highest average risk score (12.9) versus Hubballi (10.9) and Mangalore (8.9), consistent with its higher inspection failure rate.
- A rate of 80 complaints per 1,000 deliveries highlights systemic delivery-chain issues requiring process improvements beyond plant-level fixes.

Summary & Recommendations

- **North Zone Safety Priority:** With a 9.27% leakage rate — nearly double all other zones — urgent pipeline inspections and field audits are recommended.
- **Belagavi Plant Quality Programme:** Highest inspection failures (1,309) and risk score (12.9) make Belagavi the priority site for a structured quality improvement plan.
- **Batch-Level Audits:** High complaint counts in specific batches (B0080 and others in the top 10) point to production-run quality issues that can be addressed upstream.
- **Faster Complaint Resolution:** An average of 10.6 days to resolve safety-critical complaints is high. Expedited escalation workflows for High-severity cases are advised.
- **Leakage Cost Reduction:** Targeting leakage — which drives 59.48% of resolution costs — through preventive maintenance and field inspection campaigns offers the highest financial return.